

Junghyun An

SNU Undergraduate | UC San Diego Research Intern

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Education

Seoul National University

2021–2027 expected

B.S. in Mechanical Engineering; Double Major in Electrical & Computer Engineering

GPA: 4.04 / 4.30

University of California San Diego

Sep. 2026–Jan. 2027

Exchange Student

Research Interests

Multi-robot coordination; shared spatial memory; hierarchical robot learning; resource-aware robotic systems.

Research Experience

Research Intern, UC San Diego

Sep. 2026–Present

Advisor: Prof. Nikolay Atanasov

- Investigating shared 3D semantic memory for parallel task execution and cooperative bimanual manipulation.
- Developing a macro-action policy to select robot skills or wait actions from shared memory and robot information.

Undergraduate Researcher, Seoul National University

Jan. 2026–Present

Advisor: Prof. Hyun Jong Yang

- Devised contamination-aware task allocation (CATA), developed much of the mathematical formulation, and implemented NVIDIA Sionna wireless simulation for 40 AMRs completing 320 navigation tasks.
- In closed-loop simulation over ten seeds, improved mean uplink throughput by 2.6% and reduced mission completion time by 1.7% relative to closest-start allocation.

Undergraduate Researcher, SNU Robotics Laboratory

Mar. 2025–Jan. 2026

Advisor: Prof. Frank C. Park

- Formulated Neighborhood-Guided Flow Matching (NGFM), using local probabilistic PCA models of demonstrations to guide generative probability paths for robot actions.
- Built MuJoCo-based evaluation environments and demonstration datasets; implemented robosuite experiments comparing NGFM with vanilla flow matching and Diffusion Policy across demonstration budgets.
- In simulated door opening with 80 demonstrations, achieved 72.1% mean success across three seeds versus 60.3% for vanilla flow matching and 56.9% for Diffusion Policy (+11.9 and +15.2 percentage points).

Selected Manuscript

Contamination-Aware Task Allocation for Cell-Free Cloud-Robotic AMR Warehouses

Junghyun An[†], Sehyun Ryu[†], and Hyun Jong Yang.

Manuscript being revised for *IEEE Internet of Things Journal*. [†]Equal contribution.

Selected Engineering Project

FPGA CNN Accelerator, Seoul National University

Fall 2025

- Implemented key components of a VGGNet accelerator on an Artix-7 100T FPGA, mapping PE arithmetic to DSPs, configuring BRAM, and implementing AXI/APB interfaces; resolved integration and timing-closure issues.
- Identified host serial-read latency as a bottleneck; reduced the USB latency setting from 16 ms to 1 ms, cutting end-to-end execution time from 73.4 s to 49.3 s (32.9%).

Awards and Honors

1st Place (32 teams), Digital System Design Final Project, Seoul National University

Dec. 2025

3rd Place, Cryptanalysis Contest

Sep. 2023

Developed an algorithmic approach to elliptic-curve cryptanalysis as part of a three-person team.

Military Service

KATUSA Sergeant, Republic of Korea Army

Jul. 2022–Jan. 2024

Protocol Office, 2nd Infantry Division / ROK-US Combined Division.

Technical Skills

Programming: Python, C/C++, MATLAB, Verilog

Tools: PyTorch, NumPy, SciPy, MuJoCo, robosuite, NVIDIA Sionna, Git